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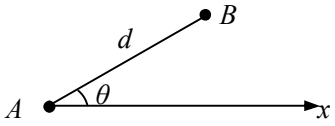
25th National High School Physics Competition: Final Problems

102008- Beijing

★ Theory

1.

In football, a player at A passes along Ax; the ball rolls on grass at speed v uniformly. A defender is at B. Let d be the A , B distance and θ the AB - Ax angle. Given $\theta < 90°$, as the ball leaves A, the defender at B starts along a straight line at speed v_p to intercept it. Ignore field boundaries. Answer the following in terms of the given parameters.



1. Find the necessary condition for interception by the defender.

2. An attacker waits on Ax, distance l_r from A. Find when the ball is not intercepted by the defender initially at B.

3. An attacker is on Ax, distance L from A. As the ball leaves A, he runs at constant speed v_r to receive it. Find when the defender initially at B does not intercept it.

2.

Ground observations give satellite motion, then the motion of spacecraft or ground objects. Both require time and spatial coordinates. To simplify analysis and calculation, ignore Earth's spin and orbit; use Earth as an inertial frame.

1. To measure satellite motion, ignore relativity. It has a radio transmitter and a precise atomic clock. Each signal contains its transmission time.

(I) Several ground stations use these data, their known positions and clocks to find its position at each instant, hence its motion. For this system,